

# Decoding Customer Value

A SQL-Driven Retention Strategy for a D2C Fashion Brand

| CUSTOMERS ANALYSED | HIGH-VALUE TIER | PROMO DEPENDENCY | AVG TRANSACTION |
|--------------------|-----------------|------------------|-----------------|
| 3,900              | 7.7%            | 43%              | \$60            |

# Executive Summary

This report presents a structured customer intelligence analysis for a direct-to-consumer (D2C) fashion brand with 3,900 customers. The central question: *Is the business building a loyal customer base, or relying on continuous promotions?*

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## ■ Promo Dependency Is High but Misdirected

43% of all purchases used a discount or promo code. Loyal customers with **low** promo reliance spend **\$60.58** on average vs \$60.04 for highly promo-reliant buyers — confirming that discounts are subsidising purchases that would have happened anyway.

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## ✓ Organic Geographic Pull Exists

Arizona, Virginia, Michigan, Tennessee, and Texas all show above-average spend with promo dependency below 40% — clear indicators of genuine brand affinity, never deliberately targeted.

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## x Three Segments Are Margin Drains

Summer Clothing, Summer Footwear, and Fall Outerwear combine high promo reliance (45%+) with 30–32% Low-value customer concentration. These are prime candidates for a promotional sunset.

## Strategic Recommendation

Phase out category-wide public discounts in the three identified sunset segments over six weeks, replacing them with behavioural triggers (48-hour cart abandonment → free shipping). Simultaneously, shift acquisition spend toward Male Gen X / Boomer customers in organic-pull states, using Accessories as the entry-point hook — the category with the highest average customer tenure (25.7 previous purchases).

# Analytical Methodology

## Dataset & Constraints

The raw dataset contains 3,900 rows with 18 columns: demographics (age, gender, location), transaction details (item, category, spend, season), behavioural signals (frequency, previous purchases, subscription status), and promotional usage. **Key constraint:** No loyalty score, no churn label, and no timestamps exist. Every analytical concept must be constructed from available variables.

## Feature Engineering

| Feature               | Logic / Formula                                                                     | Business Purpose                                    |
|-----------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------|
| dependency_score      | $(Discount\_Flag + Promo\_Flag) \div 2$                                             | Promo reliance: 0 = organic, 1 = fully incentivised |
| satisfaction_flag     | Review Rating $\geq 4.0 \rightarrow 1$ else 0                                       | High-satisfaction signal (NPS promoter zone)        |
| freq_numeric          | Frequency label $\rightarrow$ annual visits                                         | Normalises frequency strings to comparable scale    |
| composite_value_score | $40\% \text{ tenure} + 40\% \text{ freq\_rank} + 20\% \text{ spend\_rank}$          | High-end value proxy weighted by proven behaviour   |
| value_tier            | Top 25% $\rightarrow$ High; 40–75% $\rightarrow$ Med; $\geq 75\%$ $\rightarrow$ Low | Actionable tier for targeted strategy               |
| loyalty_volume        | Previous Purchases $\geq 75\text{th percentile}(\#38)$                              | Definition A: proven historical repeat behaviour    |
| loyalty_frequency     | $freq\_numeric \geq 26$ AND satisfaction $\geq 4.0$                                 | Definition B: high-frequency + satisfied buyer      |

## Loyalty Definition Validation

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### Why Volume-Based Loyalty Was Chosen

Pearson correlation with Purchase Amount: **Volume-Based = +0.0142** vs **Frequency-Based = –0.0116**. The Volume model is directionally positive and internally consistent — customers with more past purchases have *proven* they return independently of incentives. The Volume-Based flag is selected as the primary metric.

# Key Findings

## Finding 1 — Loyalty vs Promo Dependency

| Loyalty Segment | Promo Reliance             | Count | Avg Spend | Avg Tenure |
|-----------------|----------------------------|-------|-----------|------------|
| Loyal           | Low (No Incentives)        | 556   | \$60.58   | 43.9       |
| Loyal           | High (Both Discount+Promo) | 442   | \$60.04   | 44.1       |
| Non-Loyal       | Low (No Incentives)        | 1,667 | \$59.98   | 18.8       |
| Non-Loyal       | High (Both Discount+Promo) | 1,235 | \$59.01   | 19.2       |

*Insight:* Loyal customers spend equally regardless of promo usage. Discounts are not converting Non-Loyal buyers into repeat purchasers.

## Finding 2 — Geographic Opportunity

States with organic pull (avg spend > \$60, promo dependency < 40%): **Arizona, Michigan, Montana, Rhode Island, South Dakota, Tennessee**. Arizona averages \$66.55 per transaction with only 33.8% promo dependency.

## Finding 3 — Category Funnel

| Category    | Avg Tenure | Avg Spend | Funnel Role        |
|-------------|------------|-----------|--------------------|
| Accessories | 25.7       | \$59.84   | Retention Anchor ✓ |
| Footwear    | 25.2       | \$60.26   | Entry Point        |
| Clothing    | 25.2       | \$60.03   | Entry Point        |
| Outerwear   | 25.0       | \$57.17   | Entry Point        |

## Finding 4 — Promotional Sunset Candidates

| Category  | Season | % Low-Value Customers | Avg Promo Depend |
|-----------|--------|-----------------------|------------------|
| Clothing  | Summer | 32.4%                 | 45.1%            |
| Footwear  | Summer | 31.9%                 | 45.6%            |
| Outerwear | Fall   | 30.7%                 | 45.5%            |

# Retention Playbook

## 1. Promotional Sunset Plan

Three specific category × season combinations are identified as prime sunset candidates: **Summer Clothing, Summer Footwear, and Fall Outerwear**. These segments combine the highest promo reliance (≥45%) with a disproportionate share of Low-value customers (30–32%).

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### Trade-off

Removing public discounts may cause a 10–15% short-term volume drop as bargain hunters churn. However, since these customers are disproportionately low-value with shallow tenure, the impact on total LTV is minimal. Net margin per transaction will rise by an estimated 15–20%.

| Phase          | Timeline | Action                                                        | Metric to Track              |
|----------------|----------|---------------------------------------------------------------|------------------------------|
| 1. A/B Test    | Week 1–2 | 50% of traffic: no discount in sunset segments                | Net margin \$ per cohort     |
| 2. Evaluate    | Week 3–4 | If B-test margin > A-test, expand to 80%                      | 90-day repeat purchase rate  |
| 3. Full Sunset | Week 5–6 | Remove all % discounts → 48h cart-abandoners shipping segment | AOV, churn, shipping segment |

## 2. Ideal Customer Profile (ICP)

| Attribute          | Profile                                                      |
|--------------------|--------------------------------------------------------------|
| Gender             | Male                                                         |
| Age Cohort         | Gen X (36–50) and Boomers (51+)                              |
| Geography          | Arizona, Virginia, Tennessee, Michigan (organic pull states) |
| Entry Category     | Accessories (highest retention anchor — 25.7 avg tenure)     |
| Purchase Frequency | Weekly to Bi-Weekly                                          |
| Avg Spend          | \$76–\$85 per transaction                                    |
| Avg Tenure         | 40+ previous purchases                                       |
| Promo Dependency   | Low — buys without discount incentives                       |
| Payment Preference | Bank Transfer / Credit Card                                  |

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### Summary Recommendation

The business is partially reliant on promotional activity, but this dependency is concentrated in specific, low-value segments — not spread across the entire customer base. The strategic action is not to eliminate promos entirely, but to surgically remove them from segments where they are not generating long-term value, and redirect that budget into acquiring more customers who match the high-value ICP above.